

Sales Performance Dashboard

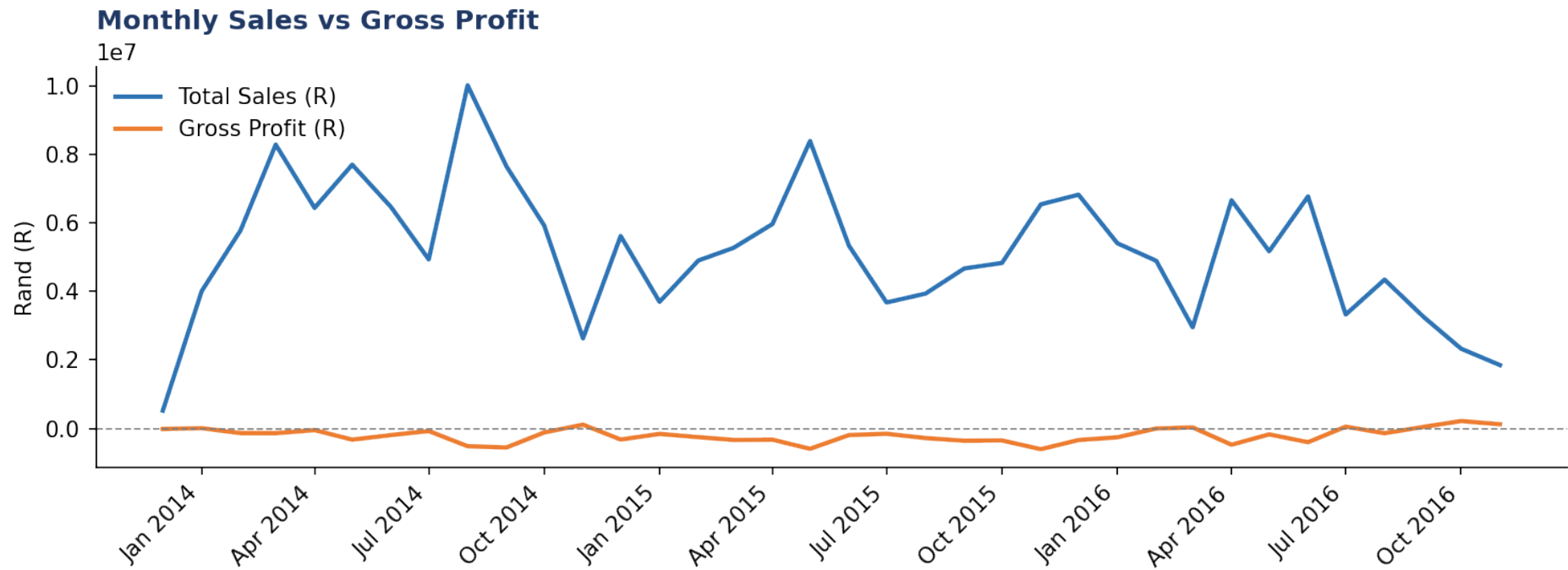
Simulated daily trading data for one product - Large retail store - 30 Dec 2013 to 16 Nov 2016 (1,053 trading days)

TOTAL SALES R186.9m Sum of daily sales, R	TOTAL QUANTITY SOLD 5,279,872 units, all days	OVERALL GROSS PROFIT % -3.81% Total GP / Total Sales	AVG UNIT PRICE (weighted) R35.40 Total Sales / Total Qty	AVG DAILY GROSS PROFIT % -0.87% Mean of daily GP%
--	--	---	---	--

Executive Summary

This simulated dataset tracks daily Sales, Cost of Sales and Quantity Sold for a single product at a large retail store across almost 3 years. Across the full period the product ran at a slight overall loss (-3.8% gross profit), driven largely by heavy, deep-discount promotional activity through 2014-2015. The good news: profitability has been recovering - 2016 is the first year with a positive average gross profit margin, and 4 of the last 5 months of data available are profitable. Demand for the product is highly price elastic (price cuts of 10-15% during promotions drove 50-170% more units sold per day), but the current depth of discounting is cutting too deep into margin - every one of the three promotional periods tested here turned a positive baseline margin into a negative one during the promotion itself. The recommendation is to test shallower promotional discounts, since elasticity this high should still generate a strong volume response at a smaller markdown.

Monthly Sales vs Gross Profit

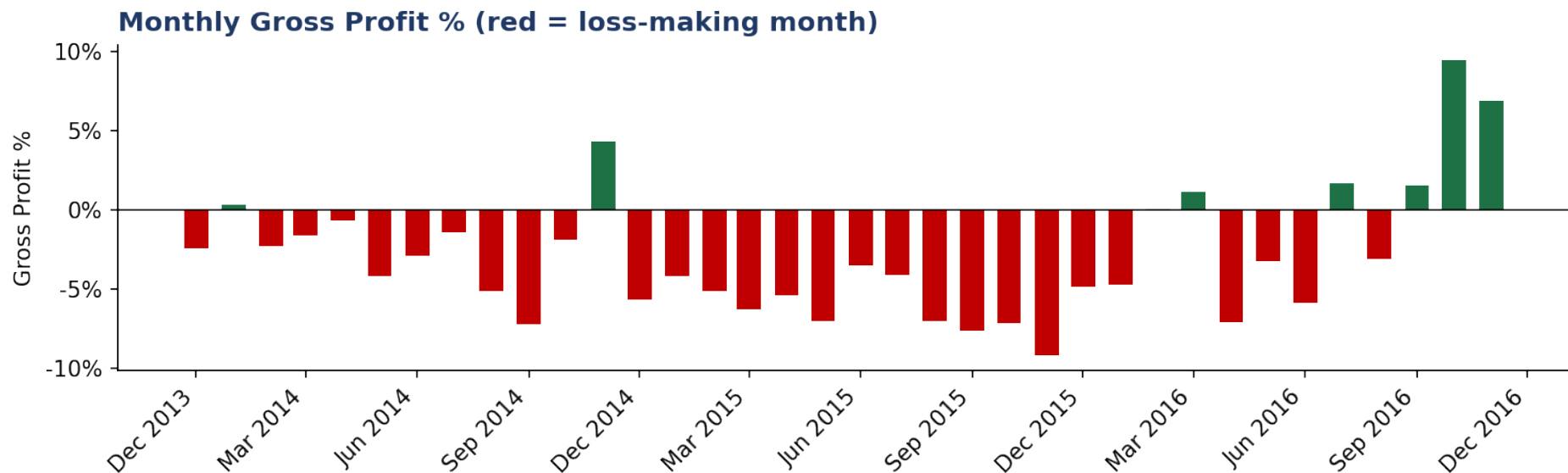


Core Metrics (Case Study Questions 1-4)

Metric	Result	Definition
1. Daily Sales Price per Unit	Ranges R30.76 - R43.81/day (full daily series in Excel workbook)	Sales / Quantity Sold, calculated per day
2. Average Unit Sales Price of Product	R37.07 (simple daily mean) R35.40 (volume-weighted)	Mean of the daily price-per-unit series; weighted version = Total Sales / Total Quantity
3. Daily % Gross Profit	Average: -0.87%/day (Overall for period: -3.81%)	(Sales - Cost of Sales) / Sales, per day
4. Daily % Gross Profit per Unit	Average: -0.87%/day (identical to #3)	(Price/unit - Cost/unit) / Price/unit -- mathematically identical to Metric 3

Note: Gross Profit % and Gross Profit % per Unit are algebraically identical -- (Sales-Cost)/Sales simplifies to (Price-Cost)/Price -- confirmed to a difference of under 1e-10 across all 1,053 days.

Monthly Gross Profit % Trend

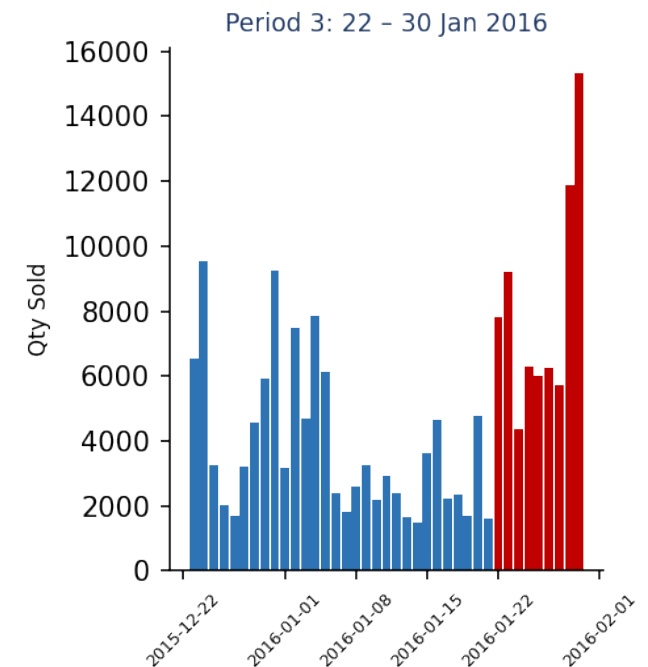
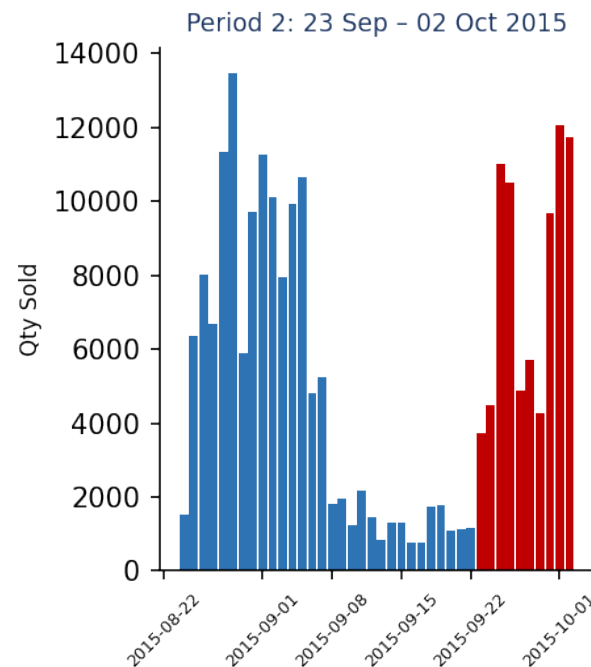
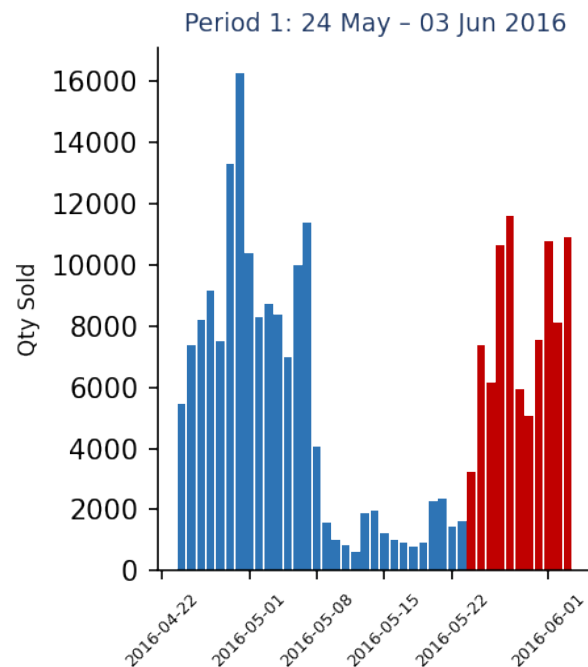


Margin was consistently negative through most of 2014-2015 and turned positive in select months from Feb 2016 onward, with sustained positive months in the second half of 2016 -- the clearest sign of improving pricing/cost discipline in the dataset.

Metric 5 - Price Elasticity of Demand During Promotions

Three promotional periods were selected (the three longest contiguous "on-promotion" runs in the data, identified from a price-drop + quantity-spike flag vs the trailing rolling average). Each is benchmarked against a 30-day, non-promotional baseline immediately preceding it.

Period	Dates	Baseline Price	Promo Price	%chg Price	Baseline Qty	Promo Qty	%chg Qty	PED	Baseline GP%	Promo GP%
1	24 May - 03 Jun 2016	R38.11	R34.35	-9.9%	5,204	7,936	+52.5%	-5.32	+2.9%	-5.9%
2	23 Sep - 02 Oct 2015	R37.95	R33.40	-12.0%	2,887	7,819	+170.8%	-14.27	+2.0%	-13.1%
3	22 - 30 Jan 2016	R40.16	R34.06	-15.2%	3,269	8,097	+147.7%	-9.72	+4.6%	-9.1%



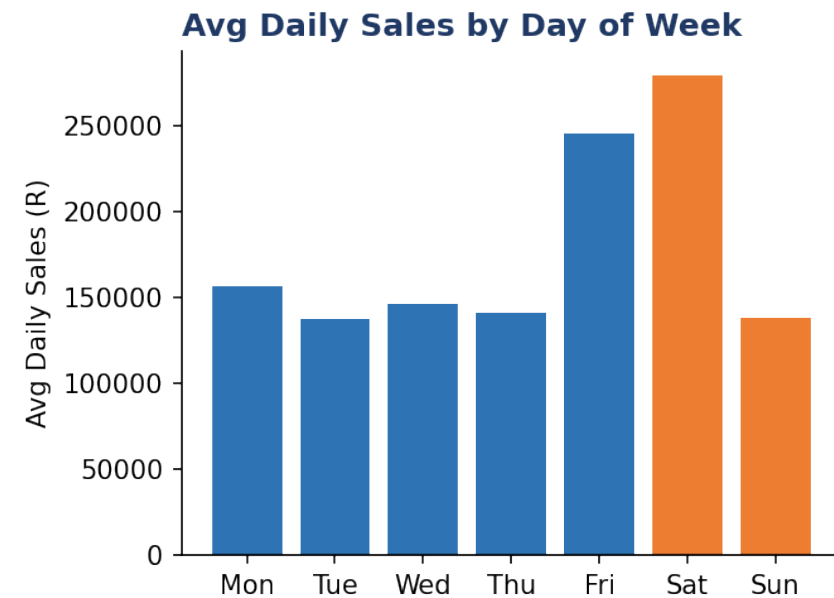
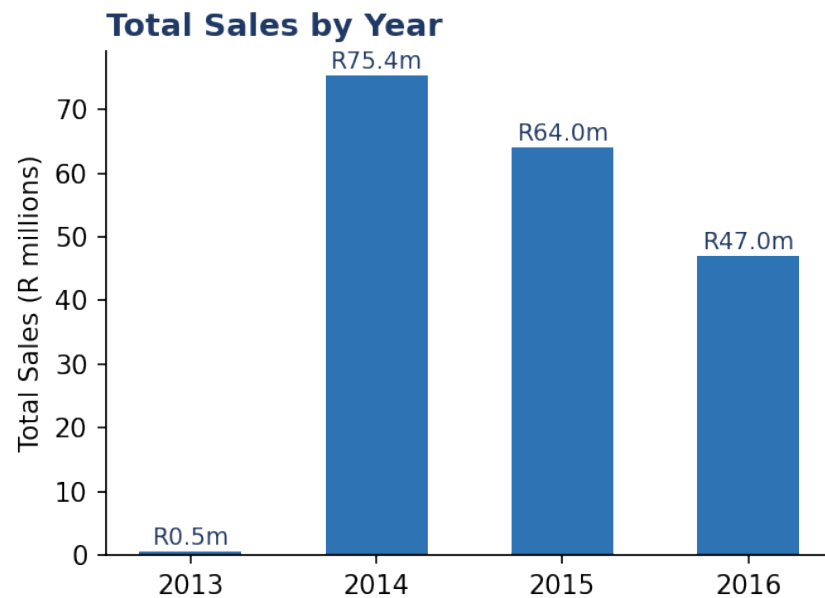
5b. Does the product perform better or worse at a promotional price?

Better on volume, worse on profit -- in its current form. All three periods confirm highly elastic demand (PED between -5.3 and -14.3): a modest price cut consistently drives a large jump in units sold, which is exactly what a retailer wants to see for a promotional lever. However, gross profit % swings from positive in the baseline period (+2.0% to +4.6%) to clearly negative during the promotion itself (-5.9% to -13.1%) in every single case. That means the discount is currently deep enough to erase the margin gained from the extra volume -- the

store is selling far more units, but making less (or losing) money doing it. Given how elastic demand is, a smaller discount (roughly half the current depth) would likely retain most of the volume uplift while protecting margin, and is the change worth testing first.

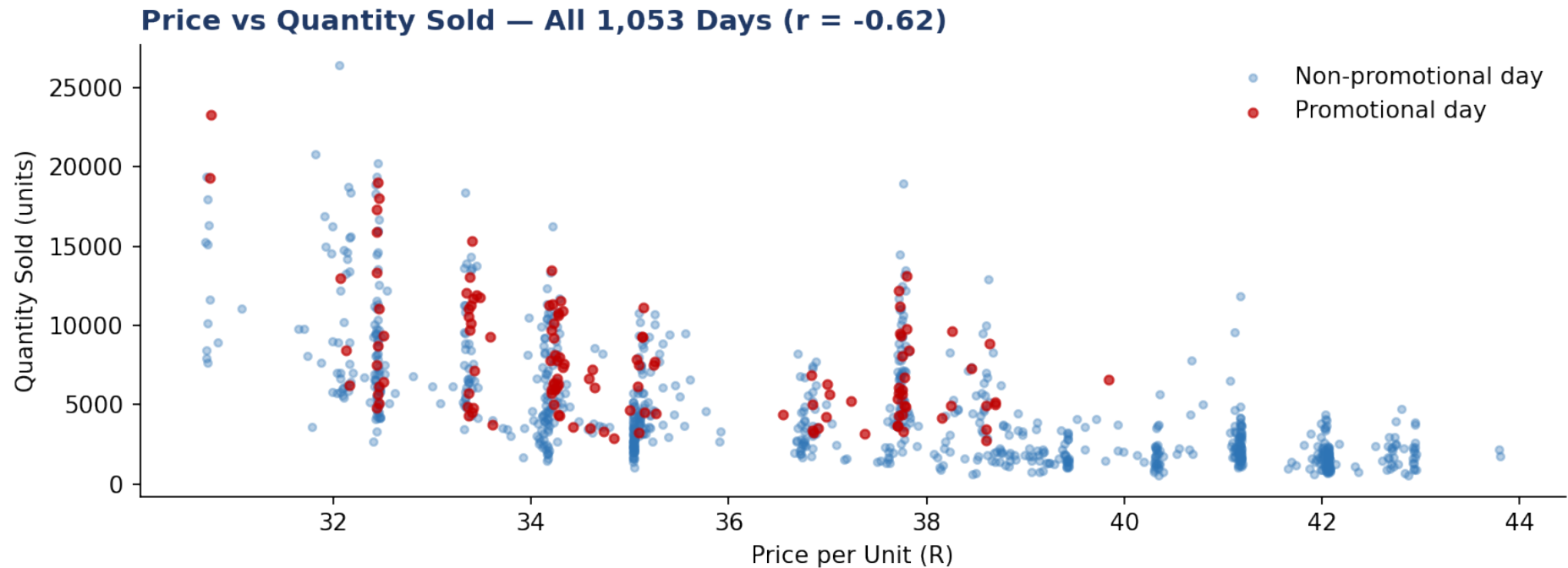
Additional Insights

Sales by Year and Seasonality



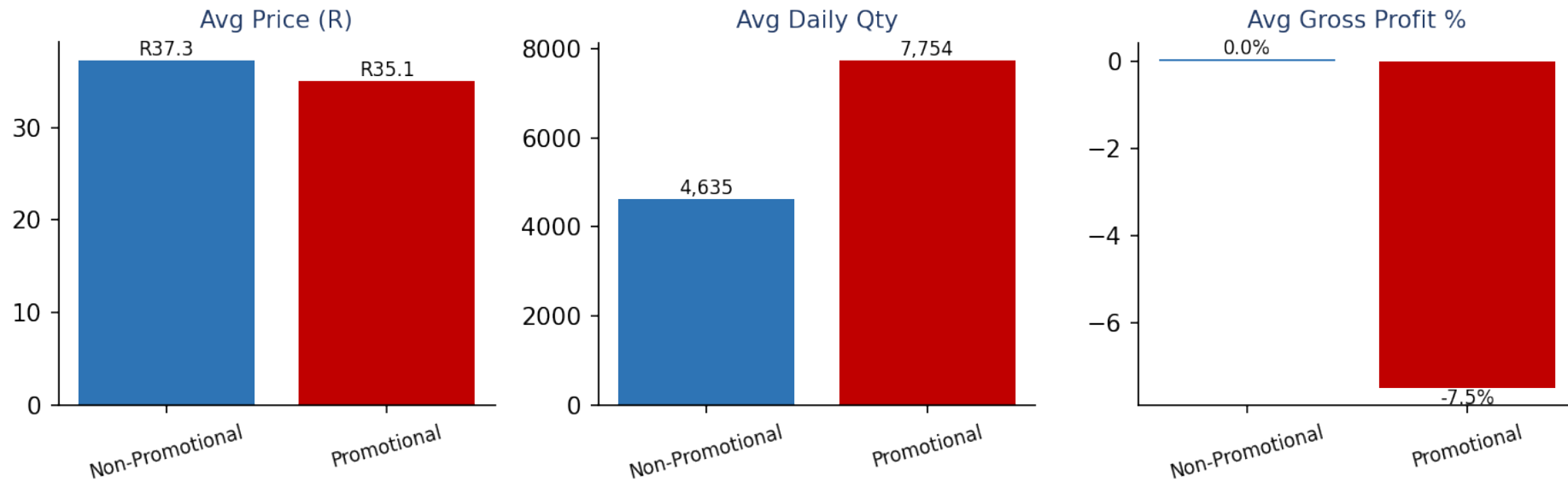
Saturdays generate the highest average daily sales (~R279k) and Sundays/Tuesdays the lowest -- weekend traffic is roughly 25% above the weekday average, but (see below) comes with thinner margins.

Price vs Quantity Relationship



Price and quantity are strongly negatively correlated ($r = \text{approx } -0.62$) across all 1,053 days -- the visible cluster of red (promotional) points sits in the lower-price / higher-quantity region, consistent with the elasticity findings above. Price also correlates positively with gross profit % ($r = \text{approx } +0.64$): higher prices protect margin, as expected.

Promotional vs Non-Promotional Days



Across the full dataset (not just the 3 tested periods), the 128 promotional days average 67% more units sold per day than the 925 non-promotional days, at a 6% lower price -- but average gross profit % flips from +0.04% (non-promo, roughly breakeven) to -7.5% (promo). This mirrors the 3-period elasticity finding at full-dataset scale and reinforces the same recommendation: current promotional pricing is too deep for its own volume gains to pay for.